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CHAPTER 10
Nonparametric Tests

WHEN AND WHY

Key Questions

- When should a nonparametric test be used?
- If nonparametric tests require fewer assumptions than parametric tests, why not always use nonparametric tests?
- Are nonparametric tests just for small samples?

In the News

Rare diseases are defined as diseases with prevalence of less than 200,000 in the United States, and those that affect fewer than 5 out of every 10,000 people in Europe. An example is pediatric stroke, which affects one in every 4,000 newborns.[a] Studies of rare diseases usually involve small samples, as there are limited numbers of patients available for enrollment, and sometimes need different statistical approaches.

Many studies consider continuous or dichotomous outcomes and involve statistical tests for means or proportions, respectively. Other studies use ordinal outcomes, where patients are classified into one of several ordered categories. For example, studies in stroke patients often use the Rankin Scale, which measures disability in patients who suffer stroke. It was developed in 1957 by Dr. John Rankin, a Scottish physiologist, as a five-level ordinal variable with scores reflecting no, slight, moderate, moderately severe, and severe disability. It was later expanded into what is today called the Modified Rankin Scale and includes seven ordered categories ranging from no symptoms at all, no significant disability, slight disability, moderate disability, moderately severe disability, severe disability, and dead.[b] Evaluating ordinal outcomes, such as the Modified Rankin Scale, also requires different statistical approaches.

Dig In

- Approximately 50% of rare diseases affect children. Are there unique challenges in studying children with rare diseases?
- Consider a clinical trial of a new intervention designed to improve disability in

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patients suffering stroke that uses the Modified Rankin Scale as the outcome. What evidence would suggest that the new intervention is effective?

- Nonparametric tests generally have lower power than parametric tests. Would you interpret results of a nonparametric test any differently than a parametric test?

LEARNING OBJECTIVES

By the end of this chapter, the reader will be able to

- Compare and contrast parametric and nonparametric tests
- Identify multiple applications where nonparametric approaches are appropriate
- Perform and interpret the Mann—Whitney U test
- Perform and interpret the Sign test and Wilcoxon Signed Rank test
- Compare and contrast the Sign test and Wilcoxon Signed Rank test
- Perform and interpret the Kruskal—Wallis test
- Identify the appropriate nonparametric hypothesis testing procedure based on type of outcome variable and number of samples

Tests for continuous outcomes focus on comparing means, while tests for dichotomous, categorical, and ordinal outcomes focus on comparing proportions. These tests are called **parametric tests** and are basedon certain assumptions for their appropriate use. For example, when running tests of hypothesis for means of continuous outcomes, all parametric tests assume that the outcome is approximately normally distributed in the population. This does not mean that the data in the observed sample follow a normal distribution, but rather that the outcome follows a normal distribution in the full population, which is not observed. For many outcomes, investigators are comfortable with the normality assumption (i.e., most of the observations are in the center of the distribution while fewer are at either extreme). It also turns out that many statistical tests are robust, which means that they maintain

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their statistical properties even when assumptions are not entirely met. Tests are robust in the presence of violations of the normality assumption when the sample size is large, based on the Central Limit Theorem. In Chapter 7 we presented a number of tests of hypothesis for continuous, dichotomous, categorical, and ordinal outcomes, and the Central Limit Theorem was presented in Chapter 5.

When the sample size is small and the distribution of the outcome is not known and cannot be assumed to be approximately normally distributed, then alternative tests called **nonparametric tests** are appropriate.

Nonparametric tests are sometimes called distribution-free tests because they are based on fewer assumptions (e.g., they do not assume that the outcome is approximately normally distributed). Parametric tests involve specific probability distributions (e.g., the normal distribution) and the tests involve estimation of the key parameters of that distribution (e.g., the mean or difference in means) from the sample data. The cost of fewer assumptions is that nonparametric tests are generally less powerful than their parametric counterparts (i.e., when the alternative is true, they may be less likely to reject H_0).

It can sometimes be difficult to assess whether a continuous outcome follows a normal distribution and thus whether a parametric or nonparametric test is appropriate. There are several statistical tests that can be used to assess whether data are likely from a normal distribution. The most popular are the Kol–mogorov–Smirnov test, the Anderson–Darling test, and the Shapiro–Wilk test.[1] Each test is essentially a goodness-of-fit test and compares observed data to quantiles of the normal (or other specified) distribution. The null hypothesis for each test is H_0: *Data follow a normal distribution* versus H_1: *Data do not follow a normal distribution.* If the test is significant (e.g.,$p < 0.05$), then data do not follow a normal distribution, and a nonparametric test is warranted. It should be noted that these tests for normality can be subject to low power. Specifically, the tests may fail to reject H_0: *Data follow a normal distribution* when in fact the data do not follow a normal distribution. Low power is a major issue when the sample size is small, which is unfortunately often when we wish to employ these tests. The most practical approach to assessing normality involves investigating the distributional form of the outcome in the sample using a histogram and augmenting that with data from other studies (if available) that may inform the assessment of the distribution of the outcome in the population.

There are some situations when it is clear that the outcome does not follow a normal distribution. These include situations when the outcome is an ordinal variable or a rank, when there are definite outliers, or when the outcome has clear limits of detection. Examples of these types of outcomes are given next.

Consider a clinical trial where study participants are asked to rate their symptom severity following 6 weeks on the assigned treatment. Symptom severity might be

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measured on a 5-point ordinal scale with the following response options: symptoms got much worse, symptoms got slightly worse, no change, symptoms improved slightly, or symptoms are much improved. Suppose there are a total of n = 20 participants in the trial, randomized to an experimental treatment or placebo, and the outcome data are distributed as shown in **Figure 10.1**. The distribution of the outcome (symptom severity) does not appear to be normal; more participants report improvement in symptoms as opposed to worsening of symptoms.

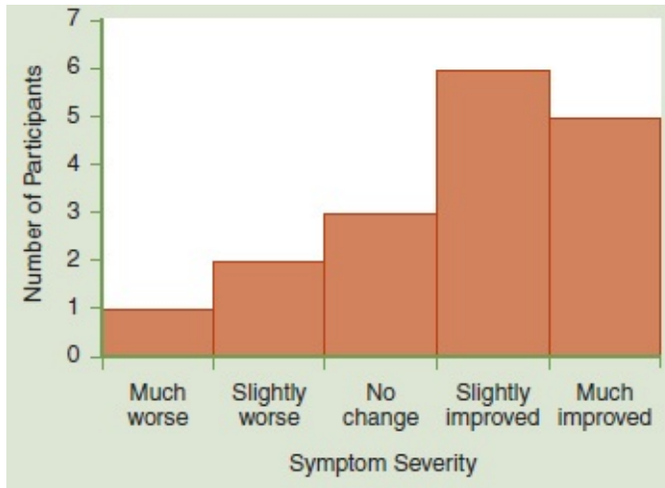

Figure 10.1 Distribution of Symptom Severity in Total Sample

Description

In some studies, the outcome is a rank. For example, in obstetrical studies an APGAR score is often used to assess the health of a newborn. The APGAR score is named after Dr. Virginia Apgar and is used to describe the condition of an infant at birth.[2] The APGAR score is based on five criteria: appearance of the skin, pulse rate, grimace (reflex or reaction to stimulation), activity (or muscle tone), and respiration. Each of the five criteria is rated as 0 (very unhealthy), 1, or 2 (healthy) based on specific clinical criteria. The APGAR score is the sum of the five component scores and ranges from 0 to 10. Infants with scores of 7 or higher are considered normal, scores of 4–6 are considered low, and scores of 0–3 are critically low. Sometimes the APGAR scores are repeated, for example, at 1 minute after birth, at 5 minutes after birth, and at 10 minutes after birth, and then are analyzed. APGAR scores generally do not follow a normal distribution; most newborns have scores of 7 or higher (normal range).

In some studies, the outcome is continuous but is subject to outliers or extreme values. For example, the number of days in the hospital following a particular surgical procedure is an outcome that is often subject to outliers. Suppose in an observational study investigators wish to assess whether there is a difference in the days patients spend in the hospital following liver transplant in for-profit versus nonprofit hospitals. Suppose days in the hospital following transplant are measured in n = 100 participants, 50 from for-profit hospitals and 50 from nonprofit hospitals;

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the data are shown in **Figure 10.2**.

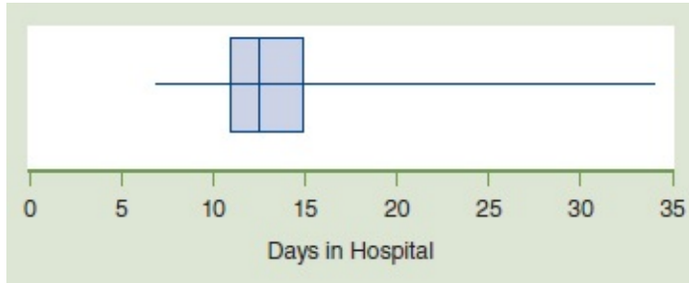

Figure 10.2 Distribution of Days in the Hospital Following Transplant

Description

Note that 75% of the participants stay at most 16 days in the hospital following a transplant, while at least one patient stayed 35 days, which would be considered an outlier. Recall that we use $Q_1 - 1.5\ (Q_3 - Q_1)$ as a lower limit and $Q_3 + 1.5(Q_3 - Q_1)$ as an upper limit to detect outliers. In **Figure 10.2**, $Q_1 = 12$ and $Q_3 = 16$; thus outliers are values below $12 - 1.5(16 - 12) = 6$ or above $16 + 1.5(16 - 12) = 22$. See Chapter 4 for more on outliers.

In some studies, the outcome is a continuous variable that is measured with some imprecision (e.g., with clear limits of detection). Some instruments or assays cannot measure presence of specific quantities above or below certain limits. For example, viral load is a measure of the amount of virus in the body and is measured as the number of copies of virus in a certain volume of blood. It can range from “not detected” or “below the limit of detection” to hundreds of millions of copies. Thus, some participants in a sample may have measures like 1,254,000 copies or 874,050 copies, and others are measured as “not detected.” If a substantial number of participants have undetectable levels, the distribution of viral load is not normally distributed.

In nonparametric tests, the hypotheses are not about population parameters (e.g., $\mu = 50$ or $\mu_1 = \mu_2$). Instead, the null hypothesis is more general. For example, when comparing two independent groups in terms of a continuous outcome, the null hypothesis in a parametric test is H_0: $\mu_1 = \mu_2$ In a nonparametric test the null hypothesis is that the two populations are equal; often this is interpreted as being equal in terms of their central tendency.

Nonparametric tests have some distinct advantages. With outcomes such as those just described, nonparametric tests may be the only way to analyze the data. Outcomes that are ordinal, ranked, subject to outliers, or measured imprecisely are difficult to analyze with parametric methods without making major assumptions about their distributions, as well as decisions about how to code some values (e.g., “not detected”). As we describe here, nonparametric tests can also be relatively simple to conduct.

Here we describe some popular nonparametric tests for continuous outcomes.

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Interested readers should see Conover[3] for a more comprehensive coverage of nonparametric tests. Before proceeding, it is worth repeating that if data are approximately normally distributed, then parametric tests are more appropriate. Parametric tests are generally more powerful and can test a wider range of alternative hypotheses. However, there are situations in which the assumptions for a parametric test are violated and a nonparametric test is more appropriate. Parametric tests are covered in detail in Chapter 7.

[a]Johns Hopkins Medicine. "Pediatric stroke." Available at: https://www.hopkinsmedicine.org/health/conditions-and-diseases/stroke/pediatric-stroke.

[b]Broderick, J.P., Adeoye, O., and Elm, J." Evolution of the Modified Rankin Scale and its use in future stroke trials." *Stroke* 2017; 48: 2007–2012.

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10.1 Introduction to Nonparametric Testing

Recall that there are various types of variables or outcomes: dichotomous outcomes have only two possible response options (usually coded “yes” and “no”); ordinal and categorical outcomes with more than two response options that are ordered and unordered, respectively; and continuous—sometimes called quantitative or measurement—variables. **Continuous variables**, in theory, take on an unlimited number of response options.

The techniques we describe here apply to outcomes that are ordinal, ranked, or continuous and are not normally distributed. Recall that continuous outcomes are quantitative measures based on a specific measurement scale (e.g., weight in pounds or height in inches). Some investigators make the distinction between continuous and interval- and ordinal-scale data. Interval data are like continuous data in that they are measured on a constant scale (i.e., the same difference exists between adjacent scale scores across the entire spectrum of scores). However, differences between interval scores are interpretable, but ratios are not. Temperature in Celsius or Fahrenheit is an example of an interval-scale outcome. The difference between 30° and 40° is the same as the difference between 70° and 80°, yet 80° is not twice as warm as 40°. Ordinal outcomes can be less specific, and the ordered categories need not be equally spaced. Symptom severity is an example of an ordinal outcome, and it is not clear whether the difference between “much worse” and “slightly worse” is the same as the difference between “no change” and “slightly improved.” Some studies use visual scales to assess participants’ self-reported signs and symptoms. Pain is often measured in this way. Participants are sometimes shown a visual scale such as that in the upper portion of **Figure 10.3** and are asked to choose the number that best represents their pain state, with 0 representing no pain and 10 representing agonizing pain. Sometimes pain scales use visual anchors, as shown in the lower portion of **Figure 10.3**.

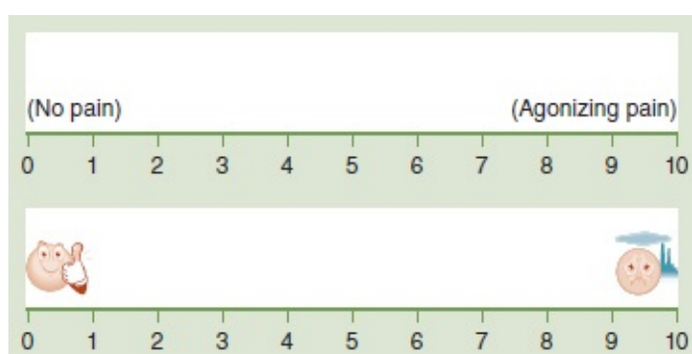

Figure 10.3 Visual Pain Scales

Description

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In the upper portion of **Figure 10.3**, although 10 is worse than 9, which is worse than 8, the difference between adjacent scores may not necessarily be the same. It is important to understand how outcomes are measured to make appropriate inferences based on statistical analysis and, in particular, not to overstate precision.

The nonparametric procedures that we describe here follow the same general procedure. The outcome variable (ordinal, interval, or continuous) is ranked from lowest to highest, and the analysis focuses on the ranks as opposed to the measured or raw values. For example, suppose we measure self-reported pain using a visual analog scale with anchors at 0 (no pain) and 10 (agonizing pain) and record the following responses in a sample of $n = 6$ participants:

7	5	9	3	0	2

The ranks, which are used to perform a nonparametric test, are assigned as follows: First the data are ordered from smallest to largest. The lowest-value score is then assigned a rank of 1, the next lowest a rank of 2, and so on. The largest value is assigned a rank of n (in this example, $n = 6$). The observed data and corresponding ranks are shown below:

Ordered Observed Data:	0	2	3	5	7	9
Ranks:	1	2	3	4	5	6

A complicating issue arises when there are ties in the sample (i.e., the same value is measured in two or more participants). For example, suppose that the following data are observed in our sample of $n = 6$:

Observed Data:	7	7	9	3	0	2

The 4th and 5th ordered values are both equal to 7. When assigning ranks, the recommended procedure is to assign the mean rank of 4.5 to each (i.e., the mean of 4 and 5), as follows:

Ordered Observed Data:	0	2	3	7	7	9
Ranks:	1	2	3	4.5	4.5	6

Using the same approach, suppose that there are three values of 7. In this case, we assign a rank of 5 (the mean of 4, 5, and 6) to the 4th, 5th, and 6th ordered values, as follows:

Ordered Observed Data:	0	2	3	7	7	7
Ranks:	1	2	3	5	5	5

Assigning the mean rank when there are ties ensures that the sum of the ranks is the same in each sample ($1 + 2 + 3 + 4 + 5 + 6 = 21$, $1 + 2 + 3 + 4.5 + 4.5 + 6 = 21$, and $1 + 2 + 3 + 5 + 5 + 5 = 21$). Using this approach, the sum of the ranks will always equal $n(n + 1)/2$. When conducting nonparametric tests, it is useful to check the sum of the ranks before proceeding with the analysis.

To conduct nonparametric tests, we follow the five-step approach for hypothesis

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testing. (See Chapter 7 for more information.)

Step 1. Set up hypotheses and select the level of significance α.
Analogous to parametric testing, the research hypothesis can be one- or two-sided (one- or two-tailed), depending on the research question of interest.

Step 2. Select the appropriate test statistic.
The test statistic is a single number that summarizes the sample information. In nonparametric tests, the observed data are converted into ranks and then the ranks are summarized into a test statistic.

Step 3. Set up the decision rule.
The decision rule is a statement that tells under what circumstances to reject the null hypothesis. In some nonparametric tests we reject H_0 if the test statistic is large, while in others we reject H_0 if the test statistic is small. These are distinguished as we describe the different tests.

Step 4. Compute the test statistic.
Here we compute the test statistic by summarizing the ranks using the test statistic identified in Step 2.

Step 5. Conclusion.
The final conclusion is made by comparing the test statistic (which is a summary of the information observed in the sample) to the decision rule. The final conclusion is either to reject the null hypothesis (because it is very unlikely to observe the sample data if the null hypothesis is true) or not to reject the null hypothesis (because the sample data are not very unlikely if the null hypothesis is true).

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10.2 Tests with Two Independent Samples

In Chapter 7, we presented techniques for testing the equality of means in two independent samples. An underlying assumption for appropriate use of the tests was that the continuous outcome was approximately normally distributed or that the samples were sufficiently large (usually $n_1 \geq 30$ and $n_2 \geq 30$) to justify their use based on the Central Limit Theorem. When the outcome is not normally distributed and the samples are small, a nonparametric test is appropriate.

A popular nonparametric test to compare outcomes between two independent groups is the Mann–Whitney *U* test. The Mann–Whitney *U* test, sometimes called the Mann–Whitney– Wilcoxon test or the Wilcoxon Rank-Sum test, is used to test whether two samples are likely to derive from the same population (i.e., that the two populations have the same shape). Some investigators interpret this test as comparing the medians between the two populations. Recall that the analogous parametric test compares the means (H_0: $\mu_1 = \mu_2$) between independent groups. See Chapter 7 for more information.

The null and two-sided research hypotheses for the nonparametric test are stated as follows:

$$H_0\text{: The two populations are equal}$$
$$H_1\text{: The two populations are not equal}$$

This test is often performed as a two-sided test, and thus the research hypothesis indicates that the populations are not equal as opposed to specifying directionality. A one-sided research hypothesis is used if interest lies in detecting a positive or negative shift in one population as compared to the other. The procedure for the test involves pooling the observations from the two samples into one combined sample, keeping track of which sample each observation comes from, and then ranking lowest to highest from 1 to $n_1 + n_2$. To illustrate the procedure, consider the following example. We address each of the five steps in the test of hypothesis but include more details to promote understanding. (In subsequent examples, the five-step approach is followed more directly.)

Example 10.1 Consider a Phase II clinical trial designed to investigate the effectiveness of a new drug to reduce symptoms of asthma in children. A total of n = 10 children are randomized to receive either the new drug or a placebo. Children are asked to record the number of episodes of shortness of breath over a 1-week period following receipt of the assigned treatment. The data are shown here:

Placebo	7	5	6	4	12
New Drug	3	6	4	2	10

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The question of interest is whether there is a difference in the number of episodes of shortness of breath over a 1-week period in children receiving the new drug compared to those receiving the placebo. By inspection, it appears that children receiving the placebo have more episodes of shortness of breath than those receiving the new drug. The question is whether this observed difference is statistically significant.

In this example, the outcome is a count, and in this sample the data do not follow a normal distribution (see **Figure 10.4**). In addition, the sample size is small ($n_1 = n_2 = 5$). Thus, a nonparametric test is appropriate. The hypotheses to be tested are given below, and we run the test at the 5% level of significance (i.e., $\alpha = 0.05$).

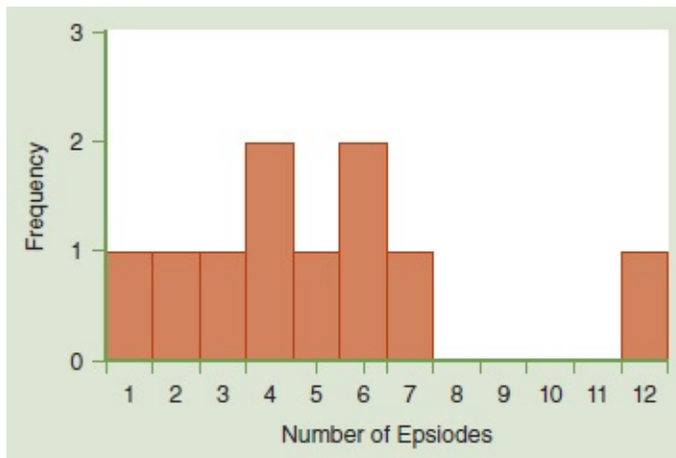

Figure 10.4 Number of Episodes of Shortness of Breath

Description

H_0: The two populations are equal
H_1: The two populations are not equal

Note that if the null hypothesis is true (i.e., the two populations are equal), we expect to see similar numbers of episodes of shortness of breath in each of the two treatment groups. Specifically, if the numbers of episodes of shortness of breath are equal in children assigned to the new drug and in those assigned to the placebo, we would expect to see some children reporting few episodes and some reporting more episodes in each of the comparison groups. This does not appear to be the case with the observed data. A test of hypothesis is needed to determine whether the observed data are evidence of a statistically significant difference in populations.

The first step is to assign ranks, and to do so we order the data from smallest to largest. This is done on the combined or total sample, pooling the data from the two treatment groups and assigning ranks from 1 to 10, as shown in **Table 10.1**. We also need to keep track of the group assignments in the total sample ($n = 10$).

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TABLE 10.1 Assigning Ranks Placebo New

		Total Sample (Ordered Smallest to Largest)		Ranks	
Placebo	New Drug	Placebo	New Drug	Placebo	New Drug
7	3		1		1
5	6		2		2
6	4		3		3
4	2	4	4	4.5	4.5
12	1	5		6	
		6	6	7.5	7.5
		7		9	
		12		10	

Description

Note that the lower ranks (e.g., 1, 2, and 3) are assigned to responses in the new drug group while the higher ranks (e.g., 9 and 10) are assigned to responses in the placebo group. Again, the goal of the test is to determine whether the observed data support a difference in the populations of responses. Recall that in parametric tests, when comparing means between two groups, we analyze the difference in the sample means relative to their variability and summarize the sample information in a test statistic. A similar approach is employed here. Specifically, we produce a test statistic based on the ranks.

First, we sum the ranks in each group. In the placebo group, the sum of the ranks is 37, and in the new drug group, the sum of the ranks is 18. Recall that the sum of the ranks will always equal $n(n + 1)/2$. As a check on our assignment of ranks, we have $n(n + 1)/2 = 10(11)/2 = 55$, which is equal to $37 + 18 = 55$.

For the test, we call the placebo group 1 and the new drug group 2 (assignment of groups 1 and 2 is arbitrary—we only need to maintain the group assignments throughout the testing procedure). We let R_1 denote the sum of the ranks in group 1 (i.e., $R_1 = 37$) and R_2 denote the sum of the ranks in group 2 (i.e., $R_2 = 18$). if the null hypothesis is true (i.e., if the two populations are equal), we expect R_1 and R_2 to be similar. In this example, the lower values (lower ranks) are clustered in the new drug group (group 2), while the higher values (higher ranks) are clustered in the placebo group (group 1). The question is whether the difference we observe in the sums of the ranks is suggestive of a difference in the populations or is simply due to chance. We now compute a test statistic to summarize the sample information and then compare that test statistic to an appropriate value from a probability distribution corresponding to our selected level of significance.

The test statistic for the Mann–Whitney U test is denoted by U and is the smaller

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of U_1 and U_2, defined as follows:

$$U_1 = n_1 n_2 + \frac{n_1(n_1+1)}{2} - R_1$$

$$U_2 = n_1 n_2 + \frac{n_2(n_2+1)}{2} - R_2$$

For this example,

$$U_1 = n_1 n_2 + \frac{n_1(n_1+1)}{2} - R_1 = 5(5) + \frac{5(6)}{2} - 37 = 3$$

$$U_2 = n_1 n_2 + \frac{n_2(n_2+1)}{2} - R_2 = 5(5) + \frac{5(6)}{2} - 18 = 22$$

Description

In our example, U = 3. Is this evidence in support of the null hypothesis or the research hypothesis?

Before we address this question, we consider the range of the test statistic U in different situations. First, consider the situation where there is complete separation between the groups. This is a situation where the data most clearly support the research hypothesis (i.e., the two populations are not equal). Specifically, suppose that all of the higher numbers of episodes of shortness of breath, and thus all of the higher ranks, are in the placebo group and all of the lower numbers of episodes of shortness of breath, and thus all of the lower ranks, are in the new-drug group and that there are no ties. Then R_1 = 6 + 7 + 8 + 9 + 10 = 40 and R_2= 1 + 2 + 3 + 4 + 5 = 15, and

$$U_1 = 5(5) + \frac{5(6)}{2} - 40 = 0 \text{ and}$$

$$U_2 = 5(5) + \frac{5(6)}{2} - 15 = 25$$

Thus, when there is clearly a difference in the populations, U = 0.

Consider a second situation where low and high scores are approximately evenly distributed in the placebo and the new-drug groups. This is a situation where the data most clearly align with the null hypothesis (i.e., the two populations are equal). Suppose that ranks of 2, 4, 6, 8, and 10 are assigned to the numbers of episodes of shortness of breath reported in the placebo group and ranks of 1, 3, 5, 7, and 9 are assigned to the numbers of episodes of shortness of breath reported in the new drug group. Then R_1 = 2 + 4 + 6 + 8 + 10 = 30 and R_2 = 1 + 3 + 5 + 7 + 9 = 25, and

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$$U_1 = 5(5) + \frac{5(6)}{2} - 30 = 10 \text{ and}$$

$$U_2 = 5(5) + \frac{5(6)}{2} - 25 = 15$$

Thus, when there is clearly no difference between populations, $U = 10$. Smaller values of U support the research hypothesis and larger values of U support the null hypothesis. In fact, for any test, the theoretical range of U is from 0 (complete separation between groups, H_0 most likely false and H_1 most likely true) to $n_1 \times n_2$ (little evidence in support of H_1). In every test, $U_1 + U_2$ is always equal to $n_1 \times n_2$. In our example, U can range from 0 to 25 and smaller values of U support the research hypothesis (i.e., we reject H_0 if U is small). The procedure for determining exactly when to reject H_0 is described next.

In every test, we must determine whether the observed U supports the null or research hypothesis. This is done following the same approach used in parametric testing. Specifically, we determine a critical value of U such that if the observed value of U is less than or equal to the critical value, we reject H_0 in favor of H_1, and if the observed value of U exceeds the critical value, we do not reject H_0. The critical value of U can be found in Table 5 in the Appendix. To determine the appropriate critical value we need the sample sizes ($n_1 = n_2 = 5$) and our two-sided level of significance ($\alpha = 0.05$). For this example the critical value is 2, and the decision rule is to reject H_0 if $U \leq 2$. We do not reject H_0 because $3 > 2$. We do not have statistically significant evidence at $\alpha = 0.05$ to show that the two populations of numbers of episodes of shortness of breath are not equal. In this example, the failure to reach statistical significance is likely due to low power. The sample data seem to suggest a difference, but the sample sizes are possibly too small to conclude that there is a statistically significant difference.

Example 10.2 A new approach to prenatal care is proposed for pregnant women living in a rural community. The new program involves in-home visits during the course of pregnancy in addition to the usual or regularly scheduled visits. A pilot randomized trial with 15 pregnant women is designed to evaluate whether women who participate in the program deliver healthier babies than women receiving usual care. The outcome is the APGAR score measured 5 minutes after birth. Recall that APGAR scores range from 0 to 10, where a score of 7 or higher is considered normal (healthy), 4–6 is low, and 0–3 is critically low. The data are shown below.

Usual Care	8	7	6	2	5	8	7	3
New Program	9	8	7	8	10	9	6	

Is there statistical evidence of a difference in APGAR scores in women receiving the new and enhanced care versus usual prenatal care? We run the test using the five-step approach.

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Step 1. Set up hypotheses and determine the level of significance.

$$H_0\text{: The two populations are equal}$$
$$H_1\text{: The two populations are not equal}$$
$$\alpha = 0.05$$

Step 2. Select the appropriate test statistic.

Because APGAR scores are not normally distributed and the samples are small (n_1 = 8 and n_2 = 7), we use the Mann–Whitney *U* test. The test statistic is U, the smaller of where R_1 and R_2 are the sums of the ranks in groups 1 and 2, respectively.

$$U_1 = n_1 n_2 + \frac{n_1(n_1+1)}{2} - R_1 \text{ and}$$
$$U_2 = n_1 n_2 + \frac{n_2(n_2+1)}{2} - R_2$$

Step 3. Set up the decision rule.

The appropriate critical value can be found in Table 5 in the Appendix. To determine the appropriate critical value we need the sample sizes (n_1 = 8 and n_2 = 7) and our two-sided level of significance (α = 0.05).

The critical value for this test is 10, and the decision rule is as follows:

$$\text{Reject } H_0 \text{ if } U \leq 10$$

Step 4. Compute the test statistic.

The first step is to assign ranks of 1 through 15 to the smallest through largest values in the total sample, as shown in **Table 10.2**. Next, we sum the ranks in each group. In the usual-care group the sum of the ranks is R_1 = 45.5 and in the new program group the sum of the ranks is R_2 = 74.5. Recall that the sum of the ranks will always equal $n(n + 1)/2$. As a check on our assignment of ranks, we have $n(n + 1)/2 = 15(16)/2 = 120$, which is equal to 45.5 + 74.5 = 120.

TABLE 10.2 Assigning Ranks

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		Total Sample (Ordered Smallest to Largest)		Ranks	
Usual Care	New Program	Usual Care	New Program	Usual Care	New Program
8	9	2		1	
7	8	3		2	
6	7	5		3	
2	8	6	6	4.5	4.5
5	10	7	7	7	7
8	9	7		7	
7	6	8	8	10.5	10.5
3		8	8	10.5	10.5
			9		13.5
			9		13.5
			10		15
				$R_1 = 45.5$	$R_2 = 74.5$

Description

We now compute U_1 and U_2, as follows:

$$U_1 = n_1 n_2 + \frac{n_1(n_1+1)}{2} - R_1 = 8(7) + \frac{8(9)}{2} - 45.5$$
$$= 46.5$$

$$U_2 = n_1 n_2 + \frac{n_2(n_2+1)}{2} - R_2 = 8(7) + \frac{7(8)}{2} - 74.5$$
$$= 9.5$$

Description

Thus, the test statistic is $U = 9.5$.

Step 5. Conclusion.

We reject H_0 because 9.5 < 10. We have statistically significant evidence at $\alpha = 0.05$ to show that the populations of APGAR scores are not equal in women receiving usual prenatal care as compared to the new program of prenatal care.

Example 10.3 A clinical trial is run to assess the effectiveness of a new antiretroviral therapy for patients with HIV. Patients are randomized to receive a standard antiretroviral therapy (usual care) or the new antiretroviral therapy and are monitored for 3 months. The primary outcome is viral load, which represents the number of HIV copies per milliliter of blood. A total of 30 participants are randomized, and the data are shown below.

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Standard Antiretroviral Therapy	7500	8000	2000	550	
	1250	1000	2250	6800	
	3400	6300	9100	970	
	1040	670	400		
New Antiretroviral Therapy	400	250	800	1400	8000
	7400	1020	6000	920	1420
	2700	4200	5200	4100	
	undetectable				

Is there statistical evidence of a difference in viral load in patients receiving the standard therapy versus the new antiretroviral therapy? We run the test using the five-step approach.

Step 1. Set up hypotheses and determine the level of significance.

$$H_0\text{: The two populations are equal}$$
$$H_1\text{: The two populations are not equal}$$
$$\alpha = 0.05$$

Step 2. Select the appropriate test statistic.

Because viral load measures are not normally distributed—generally subject to extreme values or outliers as well as clear limits of detection (e.g., "undetectable")—we use the Mann–Whitney *U* test. The test statistic is U, the smaller of

$$U_1 = n_1 n_2 + \frac{n_1(n_1+1)}{2} - R_1 \text{ and}$$

$$U_2 = n_1 n_2 + \frac{n_2(n_2+1)}{2} - R_2$$

where R_1 and R_2 are the sums of the ranks in groups 1 and 2, respectively.

Step 3. Set up the decision rule.

The appropriate critical value can be found in Table 5 in the Appendix. To determine the appropriate critical value we need the sample sizes ($n_1 = n_2 = 15$) and our two-sided level of significance α = 0.05).

The critical value for this test is 64, and the decision rule is as follows:

$$\text{Reject } H_0 \text{ if } U \leq 64$$

Step 4. Compute the test statistic.

The first step is to assign ranks of 1 through 30 to the smallest through largest values in the total sample, as shown in **Table 10.3**. Note that the "undetectable" measurement is listed first in the ordered values (smallest) and assigned a rank of 1.

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TABLE 10.3 Assigning Ranks

		Total Sample (Ordered Smallest to Largest)		Ranks	
Standard Antiretroviral	New Antiretroviral	Standard Antiretroviral	New Antiretroviral	Standard Antiretroviral	New Antiretroviral
7500	400		undetectable		1
8000	250		250		2
2000	800	400	400	3.5	3.5
550	1400	550		5	
1250	8000	670		6	
1000	7400		800		7
2250	1020		920		8
6800	6000	970		9	
3400	920	1000		10	
6300	1420		1020		11
9100	2700	1040		12	
970	4200	1250		13	
1040	5200		1400		14
670	4100		1420		15
400	undetectable	2000		16	
		2250		17	
			2700		18
		3400		19	
			4100		20
			4200		21
			5200		22
			6000		23
		6300		24	
		6800		25	
			7400		26
		7500		27	
		8000	8000	28.5	28.5
		9100		30	
				$R_1 = 245$	$R_2 = 220$

Description

Next, we sum the ranks in each group. In the standard antiretroviral therapy group, the sum of the ranks is $R_1 = 245$, and in the new antiretroviral therapy group, the sum of the ranks is $R_2 = 220$. Recall that the sum of the ranks will always equal $n(n + 1)/2$. As a check on our assignment of ranks, we have $n(n + 1)/2 = 30(31)/2 = 465$,

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which is equal to 245 + 220 = 465.

We now compute U_1 and U_2, as follows:

$$U_1 = n_1 n_2 + \frac{n_1(n_1+1)}{2} - R_1 = 15(15) + \frac{15(16)}{2} - 245$$
$$= 100$$

$$U_2 = n_1 n_2 + \frac{n_2(n_2+1)}{2} - R_2 = 15(15) + \frac{15(16)}{2} - 220$$
$$= 125$$

Description

Thus, the test statistic is $U = 100$.

Step 5. Conclusion.

We do not reject H_0 because 100 > 64. We do not have statistically significant evidence at α = 0.05 to show that the populations of viral load measures are not equal in patients receiving the standard therapy versus the new antiretroviral therapy.

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10.3 Tests with Matched Samples

We now describe nonparametric tests to compare two groups with respect to a continuous outcome when the data are collected on matched or paired samples. Here we consider the situation in which the outcome cannot be assumed to follow a normal distribution. The parametric procedure was described in detail in Chapter 7 for the situation in which the continuous outcome followed a normal distribution.

There are two popular nonparametric tests to compare outcomes between two matched or paired groups. The first is called the Sign test and the second the Wilcoxon Signed Rank test. We describe both tests here.

Recall that when data are matched or paired, we compute a difference score for each individual and analyze the difference scores. The same approach is followed in nonparametric tests. In parametric tests, the null hypothesis is that the mean difference (μ_d) is zero. In nonparametric tests, the null hypothesis is that the median difference is zero.

Example 10.4 Consider a clinical investigation to assess the effectiveness of a new drug designed to reduce repetitive behaviors in children affected with autism. If the drug is effective, children will exhibit fewer repetitive behaviors with treatment as compared with no treatment. A total of 8 children with autism enroll in the study. Each child is observed by the study psychologist for a period of 3 hours both before treatment and then again after treatment, taking the new drug for 1 week. The time that each child is engaged in repetitive behavior during each 3-hour observation period is measured. Repetitive behavior is scored on a scale of 0 to 100, and scores represent the percent of the observation time in which the child is engaged in repetitive behavior. For example, a score of 0 indicates that during the entire observation period the child did not engage in repetitive behavior, while a score of 100 indicates that the child was constantly engaged in repetitive behavior. The data are shown in **Table 10.4**. The question of interest is whether there is improvement in repetitive behavior after 1 week of treatment as compared to before treatment. Looking at the data, it appears that some children improve (e.g., Child 5 scored 80 before treatment and 20 after treatment), but some get worse (e.g., Child 3 scored 40 before treatment and 50 after treatment).

TABLE 10.4 Repetitive Behavior Before and After Treatment

Child	Before Treatment	After 1 Week of Treatment
1	85	75
2	70	50
3	40	50

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4	65	40
5	80	20
6	75	65
7	55	40
8	20	25

Because the before- and after-treatment measures are paired, we compute difference scores for each child. In this example, we subtract the assessment of repetitive behaviors measured after treatment from that measured before treatment, so that difference scores represent improvement in repetitive behavior. The question of interest is whether there is significant improvement after treatment. Again, notice that some children's repetitive behavior gets worse after treatment as compared to before treatment (and thus they have negative improvement).

The differences are shown in **Table 10.5**. In this small sample, the observed difference (or improvement) scores vary widely and are subject to extremes (e.g., the observed difference of 60 is an outlier). Thus, a nonparametric test is appropriate to test whether there is significant improvement in repetitive behavior before versus after treatment. The hypotheses are given below and the test is run at a 5% level of significance.

TABLE 10.5 Differences in Repetitive Behavior

Child	Before Treatment	After 1 Week of Treatment	Difference (Before – After)
1	85	75	10
2	70	50	20
3	40	50	–10
4	65	40	25
5	80	20	60
6	75	65	10
7	55	40	15
8	20	25	–5

Description

$$H_0\text{: The median difference is zero}$$
$$H_1\text{: The median difference is positive}$$
$$\alpha = 0.05$$

The null hypothesis, as usual, represents the situation where there is no change or no effect. In this example, the null hypothesis represents the situation where there is

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no difference in scores before versus after treatment, or no effect of treatment. If the null hypothesis is true, we expect to see some positive differences (improvement) and some negative differences (worsening). If the research hypothesis is true, we expect to see more positive differences, or more children with improvement in repetitive behavior after treatment as compared to before.

The Sign test is the simplest nonparametric test for matched or paired data. The approach is to analyze only the signs of the difference scores. The signs here are shown in **Table 10.6**. If the null hypothesis is true (i.e., if the median difference is zero), then we expect to see approximately half of the differences as positive and half of the differences as negative. If the research hypothesis is true, we expect to see more positive differences.

TABLE 10.6 Signs of the Differences

Child	Before Treatment	After 1 Week of Treatment	Difference (Before – After)	Sign
1	85	75	10	+
2	70	50	20	+
3	40	50	–10	–
4	65	40	25	+
5	80	20	60	+
6	75	65	10	+
7	55	40	15	+
8	20	25	–5	–

Description

The test statistic for the Sign test is the number of positive signs or the number of negative signs, whichever is smaller. In this example, we observe two negative and six positive signs. Is this evidence of significant improvement or simply due to chance?

We again determine whether the observed test statistic (i.e., the number of negative signs in this example) supports the null or research hypothesis. This is done following the same approach used in parametric testing. Specifically, we determine a critical value such that if the smaller of the number of positive or negative signs is less than or equal to that critical value, then we reject H_0 in favor of H_1, and if the smaller of the number of positive or negative signs is greater than the critical value, then we do not reject H_0. Notice that this is a one-sided decision rule corresponding to our one-sided research hypothesis (the two-sided situation is discussed in the next example).

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The critical value for the Sign test can be found in Table 6 in the Appendix. To determine the appropriate critical value we need the sample size, which is equal to the number of matched pairs (n = 8), and our onesided level of significance (α = 0.05). For this example the critical value is 1, and the decision rule is to reject H_0 if the smaller of the number of positive or negative signs is ≤ 1. Here we do not reject H_0 because 2 > 1. We do not have statistically significant evidence at α = 0.05 to show that there is improvement in repetitive behavior after treatment as compared to before treatment.

In our usual five-step approach, we set up a decision rule that indicates under what circumstances to reject H_0. For example, with many of our parametric tests, we reject H_0 if the observed test statistic (e.g., z or t) is more extreme than the critical value. Specifically, we reject H_0 if the test statistic is more extreme than a critical value that comes from a particular probability distribution (e.g., the standard normal distribution) based on a specified level of significance (e.g., α = 0.05). We can use the same approach for the Sign test as illustrated above, or we can consider an alternative approach such as the following.

With the Sign test we can readily compute a p-value based on our observed test statistic and compare that to our stated level of significance (e.g., α = 0.05). This is analogous to what is done when we run tests of hypothesis with a statistical computing package (e.g., using Excel or JMP). The statistical computing package computes a test statistic and a p-value, and the investigator then compares the p-value to the predetermined level of significance to draw a conclusion about the hypotheses using the following rule:

$$\text{Reject } H_0 \text{ if } p \leq \alpha$$

The test statistic for the Sign test is the smaller of the number of positive or negative signs, and it follows a binomial distribution with n = the number of subjects in the study and p = 0.5. (See Chapter 5, for more details on the binomial distribution.) In this example, n = 8 and p = 0.5. The following probabilities can be computed with either the binomial distribution formula,

$$P(x \text{ sucesses}) \frac{n!}{x!(n-x)!} p^x (1-p)^{n-x}$$

or using a statistical computing package such as Excel or JMP.

Recall that a p-value is the probability of observing a test statistic as or more extreme than that observed. We observe two negative signs. Thus, the p-value for the test is $p = P(x \leq 2)$. Using **Table 10.7**, $P(x \leq 2) = P(0) + P(1) + P(2) = 0.0039 + 0.0313 + 0.1094 = 0.1446$.

TABLE 10.7 Probabilities Based on the Binomial Distribution Model

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x = Number of Successes	$P(x$ successes$)$
0	0.0039
1	0.0313
2	0.1094
3	0.2188
4	0.2734
5	0.2188
6	0.1094
7	0.0313
8	0.0039

Because the p-value 0.1446 exceeds the level of significance ($\alpha = 0.05$), we do not have statistically significant evidence at $\alpha = 0.05$ to show that there is improvement in repetitive behaviors after treatment compared to before treatment. Notice in the table of binomial probabilities (**Table 10.7**) that we would have had to observe at most one negative sign to declare statistical significance using a 5% level of significance. Recall the critical value for our test was 1, based on Table 6 in the Appendix.

In Example 10.4 we consider a one-sided test (i.e., we hypothesize improvement after treatment). A twosided test can also be considered, in which we hypothesize a difference in repetitive behavior after treatment compared to before treatment. In Table 6 in the Appendix, we can determine a two-sided critical value and again reject H_0 if the smaller of the number of positive or negative signs is less than or equal to that of the two-sided critical value. Alternatively, we can compute a two-sided p-value. With a two-sided test, the p-value is the probability of observing many or few positive (or negative) signs. If the research hypothesis is a twosided alternative (i.e., H_1: The median difference is not zero), then the p-value is computed as $p = 2 \times P(x \leq 2)$. Notice that this is equivalent to $p = P(x \leq 2) + P(x \geq 6)$, representing the situation of few or many successes. Recall that in two-sided tests we reject the null hypothesis if the test statistic is extreme in either direction. Thus, in the Sign test, a two-sided p-value is the probability of observing few or many positive (or negative) signs. Here we observe two negative signs (and thus six positive signs). The opposite situation would be six negative signs (and thus two positive signs, as $n = 8$). The two-sided p-value is the probability of observing a test statistic as or more extreme in either direction (i.e., $P(x \leq 2) + P(x \geq 6) = 0.0039 + 0.0313 + 0.1094 + 0.1094 + 0.0313 + 0.0039 = 2 \times (0.1446) = 0.2892$).

There is a special circumstance that needs attention when implementing the Sign test that arises when one or more participants have difference scores of zero (i.e., their paired measurements are identical). If there is just one difference score of zero, some investigators drop that observation and reduce the sample size by 1 (i.e., the sample size for the binomial distribution would be $n - 1$). This is a reasonable

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approach if there is just one zero. However, if there are two or more zeros, an alternative approach is preferred. If there is an even number of zeros, we randomly assign them positive or negative signs. If there is an odd number of zeros, we randomly drop one (reduce the sample size by 1) and then randomly assign the remaining observations' positive or negative signs. The following example illustrates the approach.

Example 10.5 A new chemotherapy treatment is proposed for patients with breast cancer. Investigators are concerned with patients' ability to tolerate the treatment, so they assess quality of life both before and after receiving the new chemotherapy treatment. Quality of life (QOL) is measured on an ordinal scale and, for analysis purposes, numbers are assigned to each response category as follows: 1 = Poor, 2 = Fair, 3 = Good, 4 = Very Good, 5 = Excellent. The data are shown in **Table 10.8**. The question of interest is whether there is a difference in QOL after chemotherapy treatment as compared to before.

TABLE 10.8 Quality of Life Before and After Chemotherapy Treatment

Patient	QOL Before Chemotherapy Treatment	QOL After Before Chemotherapy Treatment
1	3	2
2	2	3
3	3	4
4	2	4
5	1	1
6	3	4
7	2	4
8	3	3
9	2	1
10	1	3
11	3	4
12	2	3

Step 1. Set up hypotheses and determine the level of significance.

$$H_0\text{: The median difference is zero}$$
$$H_1\text{: The median difference is not zero}$$
$$\alpha = 0.05$$

Step 2. Select the appropriate test statistic.

The test statistic for the Sign test is the smaller of the number of positive or negative signs.

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Step 3. Set up the decision rule.

The appropriate critical value for the Sign test can be found in Table 6 in the Appendix. To determine the appropriate critical value, we need the sample size (or number of matched pairs, n = 12) and our two-sided level of significance ($\alpha = 0.05$). The critical value for this two-sided test is 2, and the decision rule is as follows:

$$\text{Reject } H_0 \text{ if the smaller of the number of positive or negative signs is} \leq 2.$$

Step 4. Compute the test statistic.

Because the before and after-treatment measures are paired, we compute difference scores for each patient. In this example, we subtract the QOL measured before treatment from that measured after (see **Table 10.9**).

TABLE 10.9 Differences in Quality of Life Before and After Chemotherapy Treatment

Patient	QOL Before Chemotherapy Treatment	QOL After Chemotherapy Treatment	Difference (After – Before)
1	3	2	–1
2	2	3	1
3	3	4	1
4	2	4	2
5	1	1	0
6	3	4	1
7	2	4	2
8	3	3	0
9	2	1	–1
10	1	3	2
11	3	4	1
12	2	3	1

Description

We now capture the signs of the difference scores. Because there are two zeros, we randomly assign one negative sign (i.e., "–" to patient 5) and one positive sign (i.e., "+" to patient 8), as shown in **Table 10.10**.

TABLE 10.10 Signs of the Differences

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Patient	QOL Before Chemotherapy Treatment	QOL After Chemotherapy Treatment	Difference (After – Before)	Sign
1	3	2	–1	–
2	2	3	1	+
3	3	4	1	+
4	2	4	2	+
5	1	1	0	–
6	3	4	1	+
7	2	4	2	+
8	3	3	0	+
9	2	1	–1	–
10	1	3	2	+
11	3	4	1	+
12	2	3	1	+

Description

The test statistic is the number of negative signs, which is equal to 3.

Step 5. Conclusion.

We do not reject H_0 because 3 > 2. We do not have statistically significant evidence at α = 0.05 to show that there is a difference in QOL after chemotherapy treatment as compared to before.

We can also compute the *p*-value directly using the binomial distribution with n = 12 and p = 0.5. The two-sided *p*-value for the test is $p = 2 \times P(x \leq 3)$ (which is equivalent to $p = P(x \leq 3) + P(x \geq 9)$). Again, the two-sided *p*-value is the probability of observing few or many positive (or negative) signs. Here we observe three negative signs (and thus nine positive signs). The opposite situation would be nine negative signs (and thus three positive signs, as n = 12). The two-sided *p*-value is the probability of observing a test statistic as or more extreme in either direction (i.e., $P(x \leq 3) + P(x \geq 9)$). We can compute the *p*-value using the binomial formula or a statistical computing package, as follows:

$$\begin{aligned} p &= 2 \times P(x \leq 3) = 2\left(P(0) + P(1) + P(2) + P(3)\right) \\ &= 2 \times (0.0002 + 0.0029 + 0.0161 + 0.0537) \\ &= 2 \times (0.0729) = 0.1458 \end{aligned}$$

Description

Because the *p*-value 0.1458 exceeds the level of significance (α = 0.05), we do not have statistically significant evidence at α = 0.05 to show that there is a difference in QOL after chemotherapy treatment compared to before treatment.

In each of the two previous examples, we failed to show statistical significance because the *p*-value was not less than the stated level of significance. While the test

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statistic for the Sign test is easy to compute, it actually does not take much of the information in the sample data into account. All we measure is the difference in each participant's scores, and we do not account for the magnitude of the difference.

Another popular nonparametric test for matched or paired data is called the Wilcoxon Signed Rank Test. Like the Sign test, it is based on difference scores, but in addition to analyzing the signs of the differences, it also takes into account the magnitudes of the observed differences. We now use the Wilcoxon Signed Rank test to analyze the data in Example 10.4. In the following example, we address each of the five steps in the test of hypothesis but include more details to promote understanding. (In subsequent examples, the five-step approach is followed more directly.)

Example 10.6 Consider again the clinical investigation to assess the effectiveness of a new drug to reduce repetitive behaviors in children affected with autism (described in Example 10.4). A total of 8 children with autism enroll in the study, and the amount of time that each child is engaged in repetitive behavior during 3-hour observation periods is measured both before treatment and then again after taking the new drug, for a period of 1 week. The data are shown in **Table 10.11**.

TABLE 10.11 Repetitive Behavior Before and After Treatment

Child	Before Treatment	After 1 Week of Treatment
1	85	75
2	70	50
3	40	50
4	65	40
5	80	20
6	75	65
7	55	40
8	20	25

The question of interest is whether there is improvement in repetitive behavior after treatment as compared to before. Again, because the before- and after-treatment measures are paired, we compute difference scores for each child (see **Table 10.12**).

TABLE 10.12 Differences in Repetitive Beha

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Child	Before Treatment	After 1 Week of Treatment	Difference (Before – After)
1	85	75	10
2	70	50	20
3	40	50	–10
4	65	40	25
5	80	20	60
6	75	65	10
7	55	40	15
8	20	25	–5

Description

The next step is to rank the difference scores. In the ranking process, we first order the absolute values of the difference scores and assign ranks using the approach outlined previously. Specifically, we assign ranks from 1 through *n* to the smallest through largest absolute values of the difference scores, respectively, and assign the mean rank when there are ties in the absolute values of the difference scores (see **Table 10.13**).

TABLE 10.13 Ranking the Absolute Values of the Differences

Observed Differences	Ordered Absolute Values of Differences	Ranks
10	–5	1
20	10	3
–10	–10	3
25	10	3
60	15	5
10	20	6
15	25	7
–5	60	8

The final step is to attach the signs ("+" or "–") of the observed differences to each rank as shown in **Table 10.14**.

TABLE 10.14 Attaching Signs to the Ranks

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Observed Differences	Ordered Absolute Values of Difference Scores	Ranks	Signed Ranks
10	–5	1	–1
20	10	3	3
–10	–10	3	–3
25	10	3	3
60	15	5	5
10	20	6	6
15	25	7	7
–5	60	8	8

Description

Similar to the Sign test, the hypotheses for the Wilcoxon Signed Rank test concern the population median of the difference scores. The research hypothesis can be one- or two-sided. Here we consider a one-sided test.

$$H_0\text{: The median difference is zero}$$
$$H_1\text{: The median difference is positive}$$
$$\alpha = 0.05$$

The test statistic for the Wilcoxon Signed Rank test is W, defined as the smaller of W+ and W–, which are the sums of the positive and negative ranks, respectively. If the null hypothesis is true, we expect to see similar numbers of lower and higher ranks that are both positive and negative (i.e., W+ and W– would be similar). If the research hypothesis is true, we expect to see more higher and positive ranks (in this example, more children with substantial improvement in repetitive behavior after treatment as compared to before treatment, i.e., W+ much larger than W–).

In this example, W+ = 32 and W– = 4. Recall that the sum of the ranks (ignoring the signs) will always equal $n(n + 1)/2$. As a check on our assignment of ranks, we have $n(n + 1)/2 = 8(9)/2 = 36$, which is equal to 32 + 4. The test statistic is $W = 4$.

Next we must determine whether the observed test statistic W supports the null or the research hypothesis. This is done following the same approach used in parametric testing. Specifically, we determine a critical value of W such that if the observed value of W is less than or equal to the critical value, we reject H_0 in favor of H_1, and if the observed value of W exceeds the critical value, we do not reject H_0. The critical value of W can be found in Table 7 in the Appendix. To determine the appropriate one-sided critical value from Table 7 in the Appendix, we need the sample size ($n = 8$) and our one-sided level of significance ($\alpha = 0.05$). For this example, the critical value of W is 6, and the decision rule is to reject H_0 if $W \leq 6$. Thus, we reject H_0 because $4 < 6$. We have statistically significant evidence at $\alpha = 0.05$ to show that the median difference is positive (i.e., that there is significant

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improvement in repetitive behavior in children affected with autism after treatment as compared to before treatment).

Notice that when we analyzed the data presented in Example 10.4 using the Sign test, we failed to find statistical significance. However, when we use the Wilcoxon Signed Rank test (Example 10.6), we conclude that there is statistically significant evidence at $\alpha = 0.05$ to show that the median difference is positive (i.e., that there is significant improvement in repetitive behavior in children affected with autism after treatment as compared to before). The discrepant results are due to the fact that the Sign test uses very little information in the data and is a less powerful test (i.e., the test can fail to reject H_0 when in fact it is false).

Example 10.7 A study is run to evaluate the effectiveness of an exercise program in reducing systolic blood pressure in patients with prehypertension (defined as a systolic blood pressure between 120 mm Hg and 139 mm Hg or a diastolic blood pressure between 80 mm Hg and 89 mm Hg). A total of 15 patients with prehypertension enroll in the study, and their systolic blood pressures are measured. Each patient then participates in an exercise training program where they learn proper techniques and execution of a series of exercises. Patients are instructed to do the exercise program 3 times per week for 6 weeks. After 6 weeks, systolic blood pressures are again measured. The data are shown in **Table 10.15**.

TABLE 10.15 Systolic Blood Pressure Before and After Exercise Program

Patient	Systolic Blood Pressure Before Exercise Program	Systolic Blood Pressure After Exercise Program
1	125	118
2	132	134
3	138	130
4	120	124
5	125	105
6	127	130
7	136	130
8	139	132
9	131	123
10	132	128
11	135	126
12	136	140
13	128	135
14	127	126
15	130	132

The question of interest is whether there is a difference in systolic blood pressure after participating in the exercise program compared to before participating in the

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exercise program.

Step1. Set up hypotheses and determine the level of significance.

$$H_0\text{: The median difference is zero}$$
$$H_1\text{: The median difference is not zero}$$
$$\alpha = 0.05$$

Step 2. Select the appropriate test statistic.
The test statistic for the Wilcoxon Signed Rank test is W, defined as the smaller of W+ and W–, which are the sums of the positive and negative ranks, respectively.

Step 3. Set up the decision rule.
The critical value of *W* can be found in Table 7 in the Appendix. To determine the appropriate critical value from Table 7 in the Appendix, we need the sample size (*n* = 15) and our two-sided level of significance (α = 0.05). The critical value for this two-sided test is 25, and the decision rule is as follows:

$$\text{Reject } H_0 \text{ if } W \leq 25$$

Step 4. Compute the test statistic.
Because the before and after systolic blood pressure measures are paired, we compute difference scores for each patient (see **Table 10.16**).

TABLE 10.16 | Differences in Systolic Blood Pressure Before and After Exercise Program

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Patient	Systolic Blood Pressure Before Exercise Program	Systolic Blood Pressure After Exercise Program	Difference (Before – After)
1	125	118	7
2	132	134	–2
3	138	130	8
4	120	124	–4
5	125	105	20
6	127	130	–3
7	136	130	6
8	139	132	7
9	131	123	8
10	132	128	4
11	135	126	9
12	136	140	–4
13	128	135	–7
14	127	126	1
15	130	132	–2

Description

The next step is to rank the ordered absolute values of the difference scores using the approach outlined previously. Specifically, we assign ranks from 1 through *n* to the smallest through largest absolute values of the difference scores, respectively, and assign the mean rank when there are ties in the absolute values of the difference scores (see **Table 10.17**).

TABLE 10.17 Ranking the Absolute Values of the Differences

Observed Differences	Ordered Absolute Values of Differences	Ranks
7	1	1
–2	–2	2.5
8	–2	2.5
–4	–3	4
20	–4	6
–3	–4	6
6	4	6
7	6	8
8	–7	10
4	7	10
9	7	10
–4	8	12.5
–7	8	12.5
1	9	14
–2	20	15

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The final step is to attach the signs ("+" or "–") of the observed differences to each rank as shown in **Table 10.18**.

TABLE 10.18 Attaching Signs to the Ranks

Observed Differences	Ordered Absolute Values of Differences	Ranks	Signed Ranks
7	1	1	1
–2	–2	2.5	–2.5
8	–2	2.5	–2.5
–4	–3	4	–4
20	–4	6	–6
–3	–4	6	–6
6	4	6	6
7	6	8	8
8	–7	10	–10
4	7	10	10
9	7	10	10
–4	8	12.5	12.5
–7	8	12.5	12.5
1	9	14	14
–2	20	15	15

Description

In this example, W+ = 89 and W– = 31. Recall that the sum of the ranks (ignoring the signs) will always equal $n(n + 1)/2$. As a check on our assignment of ranks, we have $n(n + 1)/2 = 15(16)/2 = 120$, which is equal to 89 + 31. The test statistic is $W = 31$.

Step 5. Conclusion.

We do not reject H_0 because 31 > 25. We do not have statistically significant evidence at $\alpha = 0.05$ to show that the median difference in systolic blood pressures is not zero (i.e., that there is a significant difference in systolic blood pressures after the exercise program as compared to before the exercise program).

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10.4 Tests with More Than Two Independent Samples

Testing the equality of means in more than two independent samples can be performed using analysis of variance (ANOVA) (see Chapter 7). An underlying assumption for appropriate use of ANOVA is that the continuous outcome is approximately normally distributed or that the samples are sufficiently large (usually $n_j \geq 30$, where $j = 1, 2, ..., k$ and k denotes the number of independent comparison groups). An additional assumption for appropriate use of ANOVA is equality of variances in the k comparison groups. ANOVA is generally robust when the sample sizes are small but equal. When the outcome is not normally distributed and the samples are small, a nonparametric test is appropriate.

A popular nonparametric test to compare outcomes among more than two independent groups is the Kruskal–Wallis test. The Kruskal–Wallis test is used to compare medians among k comparison groups ($k > 2$) and is sometimes described as an ANOVA with the data replaced by their ranks. The null and research hypotheses for the Kruskal–Wallis nonparametric test are stated as follows:

$$H_0\text{: The } k \text{ population medians are equal}$$

$$H_1\text{: The } k \text{ population medians are not all equal}$$

The procedure for the test involves pooling the observations from the k samples into one combined sample, keeping track of which sample (or group) each observation comes from, and then ranking lowest to highest from 1 to N, where $N = n_1 + n_2 + ... + n_k$. To illustrate the procedure, we consider the following example, addressing each of the five steps in the test of hypothesis including more details to promote understanding. (In the subsequent example, the five-step approach is followed more directly.)

Example 10.8 A clinical study is designed to assess differences in albumin levels in adults following different low-protein diets. Albumin is measured in grams per deciliter (g/dL) of blood, and low-protein diets are often prescribed for patients with kidney disease. Clinically, albumin is also used to assess whether patients get sufficient protein in their diets. Three diets are compared, ranging from 5% to 15% protein, and the 15% protein diet represents a typical American diet. The albumin levels of participants following each diet are shown in **Table 10.19**.

TABLE 10.19 Albumin Levels in Three Different Diets

5% Protein	10% Protein	15% Protein

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3.1	3.8	4.0
2.6	4.1	5.5
2.9	2.9	5.0
	3.4	4.8
	4.2	

The question of interest is whether there is a difference in albumin levels among the three different diets. For reference, normal albumin levels are generally between 3.4 g/dL and 5.4 g/dL. By inspection, it appears that participants following the 15% protein diet have higher albumin levels than those following the 5% protein diet. The issue is whether this observed difference is statistically significant.

In this example, the outcome is continuous, but the sample sizes are small and not equal across comparison groups ($n_1 = 3$, $n_2 = 5$, $n_3 = 4$). Thus, a nonparametric test is appropriate. The hypotheses to be tested are given below and we run the test at the 5% level of significance (i.e., $\alpha = 0.05$).

$$H_0\text{: The three population medians are equal}$$
$$H_1\text{: The three population medians are not all equal}$$

To conduct the test we assign ranks. The first step in assigning ranks is to order the data from smallest to largest. This is done on the combined or total sample, pooling the data from the three comparison groups, and assigning ranks from 1 to 12, as shown in **Table 10.20**. We also need to keep track of the group assignments in the total sample ($n = 12$).

TABLE 10.20 Assigning Ranks

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			Total Sample (Ordered Smallest to Largest)			Ranks		
5% Protein	10% Protein	15% Protein	5% Protein	10% Protein	15% Protein	5% Protein	10% Protein	15% Protein
3.1	3.8	4.0	2.6			1		
2.6	4.1	5.5	2.9	2.9		2.5	2.5	
2.9	2.9	5.0	3.1			4		
	3.4	4.8		3.4			5	
	4.2			3.8			6	
					4.0			7
				4.1			8	
				4.2			9	
					4.8			10
					5.0			11
					5.5			12

Description

Notice that the lower ranks (e.g., 1, 2.5, and 4) are assigned to the 5% protein diet group, while the higher ranks (e.g., 10, 11, and 12) are assigned to the 15% protein diet group. Again, the goal of the test is to determine whether the observed data support a difference in the three population medians. Recall that in the parametric test, when comparing means among more than two groups, we analyze the differences among the sample means (mean square between treatments) relative to their within-group variability and summarize the sample information in a test statistic (*F* statistic). In the Kruskal–Wallis test we again summarize the sample information in a test statistic, one based on the ranks. Parametric tests are discussed in Chapter 7.

The test statistic for the Kruskal–Wallis test is denoted *H* and is defined as follows:

$$H = \left(\frac{12}{N(N+1)} \sum_{j=1}^{k} \frac{R_j^2}{n_j} \right) - 3(N+1)$$

where k = the number of comparison groups, N = the total sample size, n_j is the sample size in the jth group, and R_j is the sum of the ranks in the jth group.

In this example $R_1 = 7.5$, $R_2 = 30.5$, and $R_3 = 40$. Recall that the sum of the ranks will always equal $n(n + 1)/2$. As a check on our assignment of ranks, we have $n(n + 1)/2 = 12(13)/2 = 78$, which is equal to $7.5 + 30.5 + 40 = 78$.

In this example,

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$$H = \left(\frac{12}{N(N+1)} \sum_{j=1}^{k} \frac{R_j^2}{n_j} \right) - 3(N+1) =$$

$$\frac{12}{12(13)} \left(\frac{7.5^2}{3} + \frac{30.5^2}{5} + \frac{40^2}{4} \right) - 3(13) = 7.52$$

Description

In every test, we must determine whether the observed test statistic *H* supports the null or the research hypothesis. This is done following the same approach used in parametric testing. Specifically, we determine a critical value of *H* such that if the observed value of *H* is greater than or equal to the critical value, we reject H_0 in favor of H_1, and if the observed value of *H* is less than the critical value, we do not reject H_0. The critical value of *H* can be found in Table 8 in the Appendix. To determine the appropriate critical value, we need the sample sizes (n_1 = 3, n_2 = 5, and n_3 = 4) and our level of significance (α = 0.05). For this example the critical value is 5.656; thus we reject H_0because 7.52 > 5.656. We have statistically significant evidence at α = 0.05 to show that there is a difference in median albumin levels among the three different diets.

Notice that Table 8 in the Appendix contains critical values for the Kruskal–Wallis test for tests comparing three, four, or five groups with small sample sizes. If there are three or more comparison groups and five or more observations in each of the comparison groups, it can be shown that the test statistic *H* approximates a $-^2$ distribution with *df* = *k* – 1.[4] Thus, in a Kruskal–Wallis test with three or more comparison groups and five or more observations in each group, the critical value for the test can be found in Table 3 in the Appendix: Critical Values of the $-^2$ Distribution. The following example illustrates this situation.

Example 10.9 A personal trainer is interested in comparing the anaerobic thresholds of elite athletes. The anaerobic threshold is defined as the point at which the muscles cannot get more oxygen to sustain activity, or the upper limit of aerobic exercise. It is a measure that is also related to maximum heart rate. The data in **Table 10.21** are anaerobic thresholds for distance runners, distance cyclists, distance swimmers, and cross-country skiers. The question of interest is whether there is a difference in anaerobic thresholds among the different groups of elite athletes.

TABLE 10.21 Anaerobic Thresholds

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Distance Runners	Distance Cyclists	Distance Swimmers	Cross-Country Skiers
185	190	166	201
179	209	159	195
192	182	170	180
165	178	183	187
174	181	160	215

Description

Step 1. Set up hypotheses and determine the level of significance.

$$H_0\text{: The four population medians are equal}$$
$$H_1\text{: The four population medians are not all equal}$$
$$\alpha = 0.05$$

Step 2. Select the appropriate test statistic.

The test statistic for the Kruskal–Wallis test is -denoted *H* and is defined as follows:

$$H = \left(\frac{12}{N(N+1)} \sum_{j=1}^{k} \frac{R_j^2}{n_j} \right) - 3(N+1)$$

where k = the number of comparison groups, N = the total sample size, n_j is the sample size in the jth group, and R_j is the sum of the ranks in the jth group.

Step 3. Set up the decision rule.

Because there are four comparison groups and five observations in each of the comparison groups, we find the critical value in Table 3 in the Appendix for $df = k - 1 = 4 - 1 = 3$ and $\alpha = 0.05$. The critical value is 7.81, and the decision rule is to reject H_0 if $H \geq 7.81$.

Step 4. Compute the test statistic.

To conduct the test we assign ranks. The first step in assigning ranks is to order the data from smallest to largest. This is done on the combined or total sample, pooling the data from the four comparison groups, and assigning ranks from 1 to 20. We also need to keep track of the group assignments in the total sample (n = 20). **Table 10.22** shows the ordered data.

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TABLE 10.22 Ordering the Anaerobic Thresholds

				Total Sample (Ordered Smallest to Largest)			
Distance Runners	Distance Cyclists	Distance Swimmers	Cross-Country Skiers	Distance Runners	Distance Cyclists	Distance Swimmers	Cross-Country Skiers
185	190	166	201			159	
179	209	159	195			160	
192	182	170	180	165			
165	178	183	187			166	
174	181	160	215			170	
				174			
					178		
				179			
							180
					181		
					182		
						183	
				185			
							187
					190		
				192			
							195
							201
					209		
							215

Description

We next assign the ranks to the ordered values and sum the ranks in each group (**Table 10.23**).

TABLE 10.23 Assigning Ranks

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Total Sample (Ordered Smallest to Largest)				Ranks			
Distance Runners	Distance Cyclists	Distance Swimmers	Cross-Country Skiers	Distance Runners	Distance Cyclists	Distance Swimmers	Cross-Country Skiers
		159				1	
		160				2	
165				3			
		166				4	
		170				5	
174				6			
	178				7		
179				8			
			180				9
	181				10		
	182				11		
		183				12	
185				13			
			187				14
	190				15		
192				16			
			195				17
			201				18
	209				19		
			215				20
				$R_1 = 46$	$R_2 = 62$	$R_3 = 24$	$R_4 = 78$

Description

Recall that the sum of the ranks will always equal $n(n + 1)/2$. As a check on our assignment of ranks, we have $n(n + 1)/2 = 20(21)/2 = 210$, which is equal to 46 + 62 + 24 + 78 = 210.

In this example,

$$H = \left(\frac{12}{N(N+1)} \sum_{j=1}^{k} \frac{R_j^2}{n_j}) \right) - 3(N+1)$$

$$= \frac{12}{20(21)} \left(\frac{46^2}{5} + \frac{62^2}{5} + \frac{24^2}{5} + \frac{78^2}{5} \right) - 3(21) = 9.11$$

Description

Step 5. Conclusion.

Reject H_0 because 9.11 > 7.81. We have statistically significant evidence at α = 0.05 to show that there is a difference in median anaerobic thresholds among the four

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different groups of elite athletes.

Notice that in this example the anaerobic thresholds of the distance runners, cyclists, and cross-country skiers are comparable (looking only at the raw data). The distance swimmers appear to be the athletes that differ from the others in terms of anaerobic thresholds. Recall, similar to analysis of variance tests, that we reject the null hypothesis in favor of the alternative hypothesis if any two of the medians are not equal.

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10.5 Summary

In this chapter we present hypothesis testing techniques for situations with small sample sizes and outcomes that are ordinal, ranked, or continuous and cannot be assumed to be normally distributed. Nonparametric tests are based on ranks that are assigned to the ordered data. The tests involve the same five steps as parametric tests, specifying the null hypothesis and the alternative or research hypothesis, selecting and computing an appropriate test statistic, setting up a decision rule, and drawing a conclusion.

Each of the tests discussed here is summarized in **Table 10.24**.

TABLE 10.24 Summary of Key Formulas for Nonparametric Tests

Mann–Whitney *U* Test. Used to compare a continuous outcome in two independent samples.

Null Hypothesis	H_0: Two populations are equal.
Test Statistic	The test statistic is *U*, the smaller of $U_1 = n_1 n_2 + \frac{n_1(n_1+1)}{2} - R_1$ and $U_2 = n_1 n_2 + \frac{n_2(n_2+1)}{2} - R_2$ where R_1 and R_2 are the sums of the ranks in groups 1 and 2, respectively.
Decision Rule	Reject H_0 if $U \leq$ critical value from Table 5 in the Appendix.

Sign Test. Used to compare a continuous outcome in two matched or paired samples.

Null Hypothesis	H_0: Median difference is zero.
Test Statistic	The test statistic is the smaller of the number of positive or negative signs.
Decision Rule	Reject H_0 if the smaller of the number of positive or negative signs < critical value from Table 6 in the Appendix

Wilcoxon Signed Rank Test. Used to compare a continuous outcome in two matched or paired samples.

Null Hypothesis	H_0: Median difference is zero.
Test Statistic	The test statistic is *W*, defined as the smaller of *W*+ and *W*–, the sums of the positive and negative ranks of the difference scores, respectively.
Decision Rule	Reject H_0 if $W \leq$ critical value from Table 7 in the Appendix.

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Kruskal–Wallis Test. Used to compare a continuous outcome in more than two independent samples.

Null Hypothesis	H_0: *k* population medians are equal.
Test Statistic	The test statistic is *H*: $$H = \left(\frac{12}{N(N+1)} \sum_{j=1}^{k} \frac{R_j^2}{n_j} \right) - 3(N+1)$$ where *k* = the number of comparison groups, *N* = the total sample size, n_j is the sample size in the *j*th group, and R_j is the sum of the ranks in the *j*th group.
Decision Rule	Reject H_0 if H ≥ critical value from Table 8 in the Appendix.

It is important to note that nonparametric tests are subject to the same errors as parametric tests. A **Type I error** occurs when a test incorrectly rejects the null hypothesis. A **Type II error** occurs when a test fails to reject H_0 when it is false. **Power** is the probability that a test correctly rejects H_0. Nonparametric tests can be subject to low power mainly due to small sample size. Therefore, it is important to consider the possibility of a Type II error when a nonparametric test fails to reject H_0. There may be a true effect or difference, yet the nonparametric test is underpowered to detect it. For more details, interested readers should see Conover[3] and Siegel and Castellan.[4]

ILLUSTRATIVE CASE STUDY

COVID-19: Nonparametric Tests

The impact of COVID-19 on mental health is well documented, and a related implication is sleep disturbance. Sleep disturbance is a result of multiple factors, including elevated stress, worry, and anxiety; increasing screen times; and changing routines due to lockdowns and shelter in place orders.

Over 200 nursing students in Spain were assessed in January 2020 and again in April 2020 to evaluate changes in sleep quality before and during the time when the Spanish government declared a national emergency, closing the borders in an attempt to control the spread of COVID-19.[5] The primary outcome variable was sleep quality, which was measured using a validated measure, called the Pittsburg Sleep Quality Index, that assesses sleep duration, perceived quality of sleep, sleep disturbance, use of sleep medication, and so on. Each component is scored on a 5-point Likert scale and then a global or overall score is computed. Because the individual components of sleep quality were measured on a 5-point ordinal scale, the authors used nonparametric tests to test for

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differences in sleep quality measures before versus during the lockdown for COVID-19. The n = 207 nurses reported statistically significantly worse sleep quality during the lockdown, as compared to before, in every measure (e.g., subjective sleep quality [$p < 0.001$], sleep duration [$p < 0.001$], and sleep disturbance [$p = 0.017$]). Researchers also found a statistically significant difference in the overall sleep quality score ($p < 0.001$). In all measures, sleep quality was worse during the lockdown. It was also interesting to note that the nurses reported spending significantly more time in bed during the lockdown (mean 8.51 hours) as compared to before (mean 7.59 hours), $p < 0.001$, but their sleep quality was worse. The authors went on to test if there were statistically significant differences in the overall sleep quality measure in key subgroups (e.g., among males versus females, in smokers versus nonsmokers), using nonparametric tests to make the statistical comparisons. They found statistically significantly worse sleep quality during the pandemic, as compared to before, among females but not males, among first- and second-year nursing students but not among third- and fourth-year nursing students, among those reporting using alcohol, among smokers and nonsmokers, and among those with anxiety and depression and without. Poor sleep quality has been shown to be associated with increased depression, obesity, diabetes, and cardiovascular disease. This study focuses on sleep quality among a specific population, nursing students, who are already subject to substantial stress given the demands of their educational programs. It is probably true that non-nursing students also experienced a decline in sleep quality as a result of COVID-19, and the findings of this study may generalize to other groups but other studies are needed to evaluate whether these same findings persist in other populations.

For more information on sleep, sleep disorders, and associated health outcomes, please see the Centers for Disease Control and Prevention site, which has extensive data and resources on sleep and sleep disorders, and the health consequences of sleep disorders, available here: https://www.cdc.gov/sleep/publications/factsheets.html.

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